

Research Directions

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A Research Map for Quantitative Finance

From prediction

- Predict daily returns.
- Predict cross-sectional return rankings.
- Research alpha spaces for VN30 futures.

To characterization and decision

- Optimize long / hedged portfolios.
- Characterize quantitative motifs by human insight.
- Characterize motifs by inverse problems and automated discovery.

Data → Alpha → Portfolio → Scientific Explanation

Problem 1: Predict Daily Returns

Problem statement

Predict equities' and indices' daily returns:

$$r_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

for individual stocks, market indices, or sector-level instruments.

Important note

This is a common research trend, but it has a major limitation: daily return prediction is often noisy, unstable, and weak out-of-sample.

Research caution

A useful thesis should not only report model accuracy. It should also explain whether the signal survives transaction costs, regime shifts, and realistic validation.

Problem 2: Predict Daily Return Rankings

Problem statement

Instead of predicting the exact return of each stock, predict the cross-sectional ranking of future returns:

$$\text{rank}(r_{i,t+1}) \quad \text{for assets } i = 1, \dots, N.$$

Why this direction matters

This is closer to the common practice of hedge funds in global markets: build signals that rank assets, then construct long-short or market-neutral portfolios.

Typical evaluation

Information Coefficient, rank correlation, top-minus-bottom spread, portfolio return, turnover, and drawdown.

Problem 3: Optimize Long / Hedged Portfolios

Problem statement

Optimize portfolios over multiple assets, including long-only, hedged, or long-short structures:

$$\max_w \mathbb{E}[R_p] - \lambda \text{Var}(R_p)$$

under constraints on risk, exposure, liquidity, and position size.

Methods and local market example

This direction includes traditional methods such as Markowitz and Black-Litterman, as well as practical hedged structures such as long VN30 ETF and short VN30 futures.

Research value

The contribution can come from the portfolio construction layer, not only from return prediction.

Problem 4: Alpha Spaces for VN30 Futures

Problem statement

Research alpha spaces for VN30 futures as a gateway to systematic trading on a single asset.

Possible alpha dimensions

- Intraday momentum and reversal.
- Volatility expansion and compression.
- Trend-state and regime filters.
- Volume, basis, and market microstructure features.

Key challenge

The goal is not one perfect signal. The goal is to understand the space of possible signals and how they behave across regimes.

Examples of Quantitative Motifs

What is a quantitative motif?

A quantitative motif is a recurring market configuration that can be described, detected, tested, and connected to a trading decision.

Examples

- Market peaks and market bottoms.
- Patterns for a $T+2.5$ equity long strategy in the Vietnamese market.
- Short-lived price-volume structures that repeatedly appear before continuation, reversal, or volatility expansion.

Research direction

Move from informal chart observations to testable, data-driven definitions.

Problem 5: Characterize Motifs by Human Insights

Problem statement

Characterize quantitative motifs by human insights: translate experienced market observations into structured hypotheses.

Human insight layer

- What does a peak look like?
- What does accumulation look like?
- What makes a $T+2.5$ setup attractive?

Scientific layer

- Define measurable variables.
- Convert intuition into rules or labels.
- Backtest and falsify the motif.

Problem 6: Inverse Problems and Invertible Neural Networks

Problem statement

Characterize quantitative motifs by Statistical Inverse Problems and Invertible Neural Networks.

Core idea

Instead of only asking whether features predict returns, ask the inverse question:

Observed market motif $\rightarrow$ hidden market state or latent cause.

Possible contribution

Use invertible models to connect visible price-volume patterns with latent variables such as trend strength, liquidity stress, reversal pressure, or regime transition.

Problem 7: Automated Scientific Discovery

Problem statement

Characterize quantitative motifs by Automated Scientific Discovery, especially Symbolic Regression.

Core idea

Search for compact mathematical expressions that describe or detect motifs:

$$f(\text{price, volume, volatility, trend}) \approx \text{motif score.}$$

Why symbolic regression is attractive

It can produce interpretable formulas, not only black-box predictions. This makes it useful for connecting quantitative finance with scientific explanation.

Suggested Research Framing

Direction	Best framing
Return prediction	Useful but limited; requires strong validation.
Ranking prediction	More aligned with global hedge fund practice.
Portfolio optimization	Turns signals into investable decisions.
VN30 futures alpha spaces	Practical gateway to systematic trading on one asset.
Quantitative motifs	Bridge between market intuition and scientific characterization.

Overall theme

From predicting markets to understanding, testing, and systematizing market structures.